# Presentation Video — narration, v4

iGEM Claremont-CA 2026 | draft of 2026-10-25

Timecodes assume 140 words per minute; the uploaded file has a hard limit of 15:00.

Line numbers run continuously — send edits back as *line 212: …*

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## Opening

129 words ends 0:55

ON SCREEN — Team logo, a team photo, then the title card.

We're iGEM Claremont-CA, a first-year team in California, and this is Phagenomics.

We work on one question: what decides which bacteria a phage can kill?

*Pseudomonas* causes serious infections in immunocompromised patients and persistent crop disease in agriculture, and phage therapy is one of the alternatives to antibiotics people are betting on. Its bottleneck is host range. Each phage binds a specific receptor on a specific strain, and nobody can design that specificity yet, because we don't understand how a phage recognises its target in the first place.

So we didn't try to build a therapeutic. We tried to understand the recognition step that any therapeutic would depend on. We went at it four times this season. Three of those four failed, and the failures are the useful part.

## Act I.1 | The bottleneck

83 words ends 1:30

ON SCREEN — Plaque assay animation; then the same drop on two other strains, no effect.

Here is the problem in one image. We put a drop of phage onto a lawn of bacteria, and a few hours later there's a clear circle where the bacteria used to be. We move the same drop to the next plate, and nothing happens. Same phage, different strain.

Phages are fussy about what they kill, and that fussiness is the bottleneck. To treat an infection you need a phage that matches the strain in front of you, and usually there isn't one.

## Act I.2 | Adsorption and the sugar coat 66 words ends 1:59

ON SCREEN — Cross-section of a cell: membrane, wall, O-antigen, capsule. Mark which layer the phage meets first.

It gets decided at the very first step, before any DNA moves: the phage has to recognise the outside of the cell and stick to it.

And many bacteria don't present a bare protein surface. They wrap themselves in sugar — long chains called the O-antigen, and above those, in some species, a capsule. So the first thing a phage touches is a forest of sugar.

## Act I.3 | Where the project came from 79 words ends 2:33

ON SCREEN — Team photo with Prof. Libeskind-Hadas, captioned as an early conversation.

One thing belongs at the start, because it's why half of this project exists.

Early on we talked to Professor Ran Libeskind-Hadas at Claremont McKenna, one of our advisors. He suggested we reconstruct an ancestral phage —

a whole organism, inferred from its relatives. We couldn't build one. But we realised that if we scaled the idea down from an organism to a single enzyme, we could order it from a synthesis company. That is what this project became.

## Act I.4 | What we isolated

157 words ends 3:40

ON SCREEN — Sampling photo; five isolates collapsing to three strains; the circular phage genome with the tail spike outlined.

We collected soil from the California Botanic Garden on the nineteenth of May. From it we isolated five *Pseudomonas* strains, and we sequenced all five genomes — every one of them closed into a complete circle.

Those five collapse into three distinct strains: two of them the same species, the third a different one. That leaves us with two pairs of near-identical isolates, and we use those pairs as an internal control all the way through. Any difference we report between strains has to be bigger than the difference we can measure within one.

We also isolated a phage, which we call Phi-8, and it infects exactly one of the three strains. We sequenced that too: sixty-two thousand, three hundred and thirty-nine base pairs, circular, eighty genes. We found no match for it anywhere in the phage database, and essentially no genes in common with T7 or anything else characterised. The architecture is familiar. The sequence isn't.

## Act I.5 | The first clue

189 words ends 5:01

ON SCREEN — The three cassettes as gene arrows, then the presence-absence matrix of the 45 clusters with the four shared columns marked.

Then we went looking at the locus that builds the sugar coat. We anchored on the gene that flanks it in *Pseudomonas*, pulled the cassette out of all three strains, and clustered every protein in it.

We found the cassette sitting in the same slot in all three. The contents are unrelated. We counted twenty-three genes in one strain, fifteen in another, twenty-five in the third, and across all three we found forty-five gene clusters — of which only four are shared by every strain, and all four are generic sugar chemistry. No shared flippase, no shared polymerase, no shared chain-length regulator. Meanwhile the near-identical pairs inside each strain match at a hundred percent across every cassette gene. Within a strain, zero variation. Between strains, almost total.

We also found that no database could have told us any of this. There is no serotype scheme for this group of species at all.

So: three strains, three different sugar coats, one phage that kills one of them. That gave us our first hypothesis — that the sugar coat might be one of the main things setting what Phi-8 can attack.

## Cycle 1 | Design

97 words ends 5:42

ON SCREEN — DBTL bar, DESIGN lit. Lock and key.

**We designed the first cycle around the wrong assumption**, and it's worth saying why it was the reasonable one.

The textbook picture of attachment is a receptor-binding protein on the phage tail meeting a protein on the bacterial surface. Lock and key. If you assume that, the obvious move is to work out which binding protein fits which receptor — so that is what we set out to predict.

If we had known the receptor was a sugar, we would never have run this. We thought it was a protein, so we went looking for a protein.

## Cycle 1 | Build

200 words ends 7:08

ON SCREEN — BUILD lit. Funnel from 441 to 144 clusters to 114 × 63. The surrogate-host trade-off as two columns.

**What we built was a screen.**

First we needed to know what was actually on the surface, so we annotated four hundred and forty-one candidate receptors across all five isolates and collapsed them into a hundred and forty-four clusters.

Then we needed phages to compare against, which meant choosing a surrogate host — a relative of our strains that other people have already characterised phages on. We found that the closest match by receptor content had twelve phages on record, while a slightly more distant relative had two hundred and twenty-seven binding-protein entries across seventy-one phages. We took the one with the literature rather than the one with the best identity, because a receptor shared with a strain nobody has studied tells you nothing.

That left us a hundred and fourteen binding proteins against sixty-three receptors — seven thousand, one hundred and eighty-two pairs. We

folded each pair as a complex on our university's GPU cluster, and ranked them by the interface score.

Getting that to run took us three tools. ColabFold kept getting throttled by its alignment server. Boltz-2 needed a newer CUDA than the cluster had. We ended up on ESMFold, because it was the one that ran.

## Cycle 1 | Test

194 words ends 8:31

ON SCREEN — TEST lit. The ranking first so it looks reasonable, then score against protein length, then the control table. Hold 0.137 and 0.136 large.

**What we tested wasn't the screen. It was whether the screen could tell the difference.**

Six thousand, three hundred and fifty-eight of our pairs came back with a score — the rest were too long for the model — and the ranking looked perfectly reasonable. That's the trouble with this kind of output. It always looks reasonable.

So before we built anything on it, we ran nine control folds. We took three phage-receptor pairs that are known to bind, two of them with solved co-structures, and we reran each one with the receptor scrambled into nonsense, and again mismatched against the wrong phage protein.

T5's pb-five with FhuA scored zero point one three seven. The same receptor, scrambled, scored zero point one three six. The other two pairs behaved the same way. None of our real binders separated from the controls.

We also plotted score against protein length, and found a correlation of

plus nought point six. The screen was ranking proteins by size.

We assumed the tool was at fault, so we rebuilt the whole thing with real evolutionary alignments. It got worse. Under that one, the mismatched pair outscored the genuine binders.

## Cycle 1 | Learn

235 words ends 10:12

ON SCREEN — LEARN lit. Two confidence bars: on scrambling the shape score collapses 0.55 to 0.17, the interface score does not move.

**What we learned is that the method cannot answer this question at all** — and we can say exactly why.

Two tools, two control gates, both failed. One failure is about your run. Two independent failures are about the method.

The diagnostic is this. These models report two separate confidences: one for the shape of the molecule, one for how two molecules sit against each other. When we scrambled a receptor, the shape score collapsed, from zero point five five down to zero point one seven — so the model could plainly tell the sequence was nonsense. But the interface score barely moved. It had never been reading an interface at all.

That's because these models don't simulate two proteins touching. They read evolution. They look across thousands of species for positions that change in step: if a residue on one side mutates, and a residue on the other mutates to match, those two are probably in contact.

And that signal only exists where two proteins have a shared history of co-adaptation. A phage protein and a bacterial protein have never co-

adapted as a pair, so there is no history to read, and the interface score returns noise.

So this was never a result about our proteins. It's a property of the method, and more compute doesn't fix it. Nine control folds decided the fate of seven thousand, and we never ran the real screen.

## Cycle 2 | Design and Build

153 words ends 11:18

ON SCREEN — The classifier over 80 proteins; then the rung-period idea drawn on a coiled prism.

**For the second cycle we changed the question.** If we couldn't ask what the tail spike binds, we could ask what it *is*.

**We built that on two readouts that don't share a method.** First we ran an open-source sequence classifier over all eighty Phi-8 proteins to pick out the candidates. Then we measured geometry directly. A right-handed beta-helix is a chain coiled into a long prism, with its coils hydrogen-bonded in register — so if you take every backbone hydrogen bond, measure how far apart in sequence its two partners sit, and histogram those offsets, a beta-helix peaks at its coil period, somewhere between sixteen and twenty-six residues. An alpha-helix peaks at four. A beta-sandwich has no peak at all.

We chose that test rather than fell back on it. On a genome with no close relatives, a database can't contain the answer — and homology annotation had already missed something here.

## Cycle 2 | Test and Learn

257 words ends 13:08

ON SCREEN — The seven-protein verdict table, five of them turned down. Then the beta-helix render and a sugar-cutting animation.

**What we tested was eight predicted structures covering seven proteins** — our annotation splits the tail spike into two pieces that our PI's merges into one — and we classified each from its own backbone geometry.

Five of the seven came back negative, and the numbers they returned aren't noise: an alpha-helical fibre lands at six, right about where an alpha-helix should. The method says no, which is what makes it worth believing when it says yes.

It said yes twice. CDS 11, the tail spike, at classifier confidence nought point nine nine seven and a coil period of sixteen to seventeen — a right-handed beta-helix. And CDS 24, which the annotation had filed as a hypothetical protein, at nought point nine nine eight on a different period: a second depolymerase that homology-based annotation had missed entirely on this novel genome.

**What we learned from that fold is what the receptor has to be.** That shape is a signature. Enzymes that chew through long sugar chains look like this, because a sugar chain is a thread and you need something that holds it straight while you cut. You can recognise a wood plane by its shape without being told what it cuts.

So the tail spike isn't a key. It's a blade — and the lock we had spent months looking for doesn't exist.

We weren't the only ones reading it that way. Our classifier flagged it, and our PI reached the same conclusion from his own structural search. Between us, we turned it into a hypothesis.

## Cycle 2 | The hypothesis

44 words ends 13:27

ON SCREEN — One-sentence hypothesis box on white, held. Credited jointly with the PI.

In bacteria that wrap themselves in sugar, the ceiling on host range is set by what the depolymerase can digest — not by which protein the tail binds. And the narrow specificity we see today is a narrowed-down version of something broader further back.

## Cycle 2 | The prediction, and its test

135 words ends 14:24

ON SCREEN — The 18-host matrix. Put all three caveats on screen with it: no no-phage control, one isolate unscored, two readouts never recorded.

That hypothesis makes a prediction we can falsify: if the tail spike reads Strain 2's sugar, then Phi-8 should fail on the other two strains.

**We tested that in July.** We spotted eighteen host isolates against two pooled lysates, and the Strain 1 and Strain 3 representatives came back negative against both. That's the direction the prediction makes.

We're not calling it confirmed, and the reasons come from our own notebook rather than from a reviewer. We ran the assay with no no-phage control — there's a margin note on the page that just says *need control*. We didn't score one of the isolates. And two follow-up readouts that

would have settled it were never written down. The result is consistent with the prediction, and nothing contradicts it, but we haven't done the clean experiment.

## Cycle 3 | Line one: the enzyme that reads the receptor

115 words ends 15:14

ON SCREEN — Domain map of CDS 11, and the Registry card for the submitted fragment.

**For the third cycle we built the enzymes.** If the depolymerase sets the ceiling, then the test is whether that domain, on its own, can cut the sugar.

We designed the construct from the structure. CDS 11 is nine hundred and forty-one residues, and most of it isn't the enzyme: the first hundred and ten anchor it to the virion, and the next two hundred are a shaft that moves between every model we ran. The rigid trimeric body — from residue three hundred to the end, agreeing between models to within one ångström — is the part that carries the chemistry. That's the fragment we expressed, and the one we submitted to the Registry.

## Cycle 3 | Line two: why the ancestors are a different protein

232 words ends 16:53

ON SCREEN — The judge will ask this. CDS 11 and CDS 12 side by side, captioned a separate protein, not a domain.

**The ancestors, though, are a different protein, and we should be**

**clear about that.**

Phi-8 also carries CDS 12, a glycoside hydrolase from family five. It's a TIM barrel, not a beta-helix, and its closest structural match is an endoglucanase caught with a short cellulose chain in its active site. It is a separate protein — not a domain of the tail spike — and we have not established what it does inside the phage.

So why did we reconstruct ancestors of that one, rather than of the tail spike we've just spent five minutes on? Three reasons, and none of them is that we got confused. CDS 12 has a thousand homologues in the databases, which is what you need to build a tree and reconstruct along it — the tail spike is novel enough that we had no such tree. CDS 12 is a third of the size, so we judged it the far better bet to express. And we were working inside a sponsored synthesis allowance, where four short constructs were affordable and four long ones were not.

So our ancestral panel tests the shape of the hypothesis — whether an ancestral glycoside hydrolase has a broader substrate range than its descendant — on an enzyme we could actually build, rather than on the one that reads the receptor. That's a real limitation, and it's ours, not a reviewer's.

## Cycle 3 | Build

171 words ends 18:06

ON SCREEN — Tree collapsing to three nodes; Registry cards for the six parts, published and CC0.

**What we built was six constructs.**

We worked out the direction with our PI, and he ran the alignment, the tree and the reconstruction on sequences we prepared. He reconstructed every node on the path from CDS 12 back to the root, and we picked three of them to order as DNA.

All six went into pET-28a with an N-terminal His-tag and were synthesised by Twist across two orders, and all six are now published basic parts on the iGEM Registry under a CC0 licence. Three of the six are those reconstructed ancestors — protein sequences that exist nowhere in nature. We think any team working on an enzyme family could take the same route to new parts, so we wrote the method out in full.

One warning we'd pass on, because it cost us. Our first order was codon-optimised and our second wasn't, and the second set carries a mammalian-like codon bias — which in an *E. coli* host is the first thing to suspect when a band doesn't appear.

## Cycle 3 | What the model said, and its own control

151 words ends 19:11

ON SCREEN — ZymCTRL perplexity along the path, with the 21-of-24 control laid over it. Caption: perplexity is not activity.

**Before any of them reached a bench, we asked a model what it thought.** We scored the reconstructions with a protein language model against twenty-four glycoside-hydrolase classes, and the one it likes best is endo-one-four-beta-glucanase — the same chemistry the structure points at. And along the path to the root the score improves steadily, which, read on its own, is our hypothesis coming true.

Except that it doesn't survive its own control. A reconstructed ancestor is consensus-like by construction, so it scores better under almost any label — and it does. The score improves in twenty-one of the twenty-four classes. Direction alone proves nothing. What survives is that the improvement is largest under the two labels the structure points at, and even that is less clean than a single number would suggest.

And perplexity isn't activity. Nothing in that analysis measures substrate range, which is the whole point of the panel.

## Cycle 3 | Test

157 words ends 20:18

ON SCREEN — Four test rounds: the 9-15 gels, the 9-29 band, the 10-01 elution that worked and the one 29 kDa short, the 10-10 column that did not.

**We tested expression four times, and the rounds didn't line up neatly.**

On September the fifteenth we ran two gels across four constructs, induced against uninduced, and saw no detectable induction on either. The tail spike lane was essentially blank, which points at lysis or loading rather than at expression.

On the twenty-ninth, CDS 12 gave us the first induction-dependent band at the right size.

On October the first we ran nickel columns on six constructs, and one of them worked: the first ancestor eluted at its predicted mass, the only construct that did. The tail spike eluted something too, but twenty-nine kilodaltons short of what it should be.

On October the tenth we ran the same column on a second ancestor, and

both elutions came off blank, while the load and the flow-through carried the same band. We read that as a purification we had botched, and we were about to spend another week on it.

## Cycle 3 | Learn: an outside view changed a result 263 words ends 22:11

ON SCREEN — The strongest human-practices beat. Tolstorukov's email, then the 10-19 dot blot with 2.9× marked.

**What stopped us wasn't an experiment. It was a conversation.**

On October the thirteenth we took the problem to Professor Ilya Tolstorukov at Keck Graduate Institute, who works on protein expression. He wrote back before we met to say he wasn't sure he was the right person — expression is his field, purification isn't — and he met us anyway. That turned out to be the point.

He changed the question. Not *why is nothing eluting*, but *is there anything there to elute*. He told us to check whether the protein was being made at all, with a dot blot against an anti-His antibody, before we touched the column again. He also told us to run uninduced controls, and to standardise how much material goes onto a membrane before we compare spots on it.

Six days later we ran that blot, and the flow-through from the second ancestor read two point nine times the negative control.

**So what we learned is that the protein had been there all along.** It was made, it came off in the flow-through, and the run had failed at

capture — not at expression. A gel can't tell those apart: no expression, insoluble protein, and a column that didn't bind all look identical. One cheap assay separates them, and it pointed our fix at the column instead of at the construct.

We had spent a week debugging the last step of a workflow without checking that the first step worked, because the last step was the one visibly failing. That's the generalisable mistake, and one meeting fixed it.

## Cycle 3 | The activity assay, and what it carries 177 words ends 23:27

ON SCREEN — The 10-16 spot test and colony counts against the PBS control. All three caveats on screen.

**Then we ran the assay the whole project points at:** does an enzyme, on its own, do anything to the sugar?

We read it on October the sixteenth. We saw no visible lysis on any plate — including Phi-8 on its own hosts, which tells us the assay wasn't reading clearing at all. But when we counted the colonies inside the drops, we got a number. Under the first ancestor, across seven strains, colony counts fall to a median of twelve percent of the plate's PBS control. Phi-8's own median is zero.

Our PI reads that suppression as enzyme activity. We'd want more before we call it that ourselves, and there are two specific gaps. We didn't run the control that separates protein activity from phage carried over into

the preparation. And the dilutions we used in that spot test later read at or below the detection floor of our own dot blot — so we never actually showed there was protein in those drops.

It's the strongest activity signal we have, and it is one plate.

## Cycle 3 | What we still owe

50 words ends 23:48

ON SCREEN — Honest list.

And one gap runs through all of it. We have now run eight protein assays — seven gels and a dot blot — and not one of them carries a lane of untransformed BL21. We designed every round around a different variable instead. That control is on the bench now.

## Cycle 4 | Design and Test

203 words ends 25:15

ON SCREEN — Picks from the plaque zone, sequencing, 70 candidate sites down to one, F126L in a 30 kb capsule cluster.

**For the fourth cycle we stopped asking the phage, and asked the bacteria.** Three cycles in, every guess about the receptor had come from the phage's side.

In July we picked colonies from inside the plaque zone — cells still growing where the phage should have killed them — and in October we sequenced one from each host against its parent. There's no structure prediction anywhere in this loop.

We found that both picks were the parent strain, intact, with no phage

sequence in them, so neither had simply become a lysogen. A naive assembly comparison gave us sixteen to twenty-one differing sites, which means nothing on its own, because two sequencing runs of the same clone already differ by more than that. So we took every candidate back to the raw reads: seventy sites, checked against the parent's reads and against an independent read set with completely different chemistry.

Exactly one survived. We found a missense change in a Stealth-family sugar phosphotransferase, carried by two hundred and sixty-three of two hundred and sixty-four mutant reads and by none of the parent's. It sits inside a thirty-kilobase cluster that builds and exports capsular polysaccharide. The other pick has no confirmed mutation at all.

## Cycle 4 | Learn

115 words ends 26:05

ON SCREEN — A capsule hiding an intact O-antigen: how both conclusions hold at once.

**What we learned disagrees with Cycle 2 — which is the whole reason we did it this way round.** Cycle 2 told us the receptor is a sugar. Cycle 4 agrees it's a sugar, and disagrees about which one: it points at the capsule, with the O-antigen cassette intact.

Both can be true at once, because a capsule sits on top and can hide an intact O-antigen underneath it. We don't know which surface Phi-8 reads, so wherever the distinction matters we say *surface polysaccharide* and leave it there.

And the honest gap: we never confirmed those colonies were resistant.

They were colonies in a plaque zone, which is suggestive and isn't the same thing.

## What we're leaving behind

215 words ends 27:37

ON SCREEN — Tool output, the GitLab repository, the six parts, the negative result.

We're leaving four things behind, and one of them is a failure.

We packaged two tools, MIT licensed, on iGEM's GitLab. One does ladder calibration and densitometry on a gel photograph — we re-typed that code three times across three gel days before admitting we should generalise it. The other is the fold test from Cycle 2: it takes a single predicted structure and returns a call with no database behind it, and it also tells a terminal repeat from a circular join by the shape of the read coverage. That one changed our own answer. This wiki used to say our genome was linear; the check said circular, and circular is what we now report. Two more tools are staged with porting notes.

We registered six parts, three of them sequences that never existed.

We left a benchmark: on a genome with no close relatives, a machine-learning classifier found a depolymerase that homology-based annotation had filed as hypothetical. For novel phage genomes, we'd recommend running both and reconciling the disagreements.

And we published the screen that failed, with the controls that caught it and the reason it could never have worked. Don't use interface scores to screen phage-host protein pairs, and run a positive control with a solved

co-structure before you scale to thousands.

## Coda | The garden

222 words ends 29:12

ON SCREEN — Interview clips, at least 20 seconds. Credit both hosts and note permission.

Two days before our wiki froze, we went back to the garden the soil came from, and asked the people who manage it whether a treatment as strain-specific as ours would be any use to them.

We found that most of what we had assumed was wrong. We expected to argue for biological control against a chemical default, and we didn't have to — they already suppress soil fungi by mulching. We expected the objection to be ecological risk. It wasn't; it was whether the thing works. Copper is the only chemistry they have for bacterial disease, it goes onto every blossom every four or five days, and they have eighty-six acres and eight staff. The objection is arithmetic.

And here's the finding that costs us most. They have never had a sample come back positive for *Pseudomonas*. We isolated a soil phage and its environmental host — not a treatment for a disease this stakeholder actually has.

What they do need is something for bacteria inside the vascular system, where nothing they have reaches. Our strain doesn't solve that. But the question we worked on — what a tail spike reads on a cell surface, and how far you can predict that from a genome — is the one you'd have to answer first. The strain is local. The question isn't.

## Coda | Safety

134 words ends 30:09

ON SCREEN — The IBC letter, 0 VFDB / 0 CARD, and the open condition quoted.

On safety: our institutional biosafety committee gave us conditional approval with one condition still open, and that condition is a question about host range — the subject of this entire project. They asked us to show our reasoning that the work won't alter host range, virulence, or antibiotic resistance. We don't have full approval, and we're writing the revision now.

What we can say is that Phi-8 carries no virulence factors and no resistance genes across all eighty coding sequences, and that our constructs are protein fragments in laboratory *E. coli*. Nothing here is a modified phage. But understanding what sets the ceiling on host range is the same as knowing what would have to change to raise it, so we treat this as dual-use rather than arguing our way to a clean no.

## Coda | Where we are

243 words ends 31:54

ON SCREEN — Three columns fading in: Established / Partly tested / Open.

So let us say plainly where we are.

We have established five bacterial genomes and a novel circular phage genome; three unrelated sugar cassettes sitting in the same locus, with no serotype scheme in existence that could have told us; a tail spike that is a sugar-cutting enzyme by two independent methods; a controlled negative result about interface prediction; and one resistance mutation

verified in the reads.

We have partly tested two things. Phi-8 should not infect the other two strains — our July assay pointed the right way, and carried no control. And the enzyme panel: one ancestor purified at its predicted mass, and colony counts under it fall to twelve percent of control, which our PI reads as activity and which we won't call activity until we run the control that separates it from carried-over phage.

And we have left plenty open: a clean expression round with the lane we keep not running; whether the tail spike fragment is even full length; which sugar layer the phage reads; whether Phi-8 is temperate; and depositing the genome, which we haven't done.

Four cycles, three of them wrong, and each one wrong in a way that told us where to go next. If we could pass on one thing, it's this: put the cheap control in front of the expensive experiment. Nine folds told us what seven thousand couldn't, and one afternoon's dot blot told us what a week on a column hadn't.

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4466 spoken words. With the act cards and a 30-second credit card, the finished file runs about 32:54 at 140 words per minute.